

(Fully) Homomorphic Cryptographic Primitives Within the p -Adic Model of Encryption

Jaime Punyed, Addison Julistiono, Juan Reyes

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Abstract

We explore the idea of using the p -adic number system to create homomorphic and fully homomorphic encryption schemes as described by [1]. We present all the possible homomorphic ciphers on arithmetic operations over the p -adic numbers that can be achieved and show that fully homomorphic ciphers can only be created under certain conditions. We also explore some applications and implementations of these new cryptographic primitives.

1 Introduction

Cloud computing services such as Google Cloud, Microsoft Azure, and Amazon's AWS have allowed consumers to scale their applications without the need to establish their own on-premise machines by allowing them to host the applications on third-party servers. However, there is an inherent privacy risk when we allow these services to gain access to our data to do the computations. For example, say a client wants to do some computation g on a plaintext w on the cloud service. Therefore, the privacy concern is that the client would need the service to have access to w , potentially leaking the data stored in w to the service provider.

One solution to the privacy concern is the homomorphic ciphers, which allow for certain computations to be done on encrypted messages. In our example, say that the homomorphic

cipher is a function c that maps plaintext to ciphertext, then the client would instead send $C = c(w)$ to the service, and have the service compute $g(C)$. Then the client decipheres the result $g(C)$ by computing $c^{-1}(g(C)) = c^{-1}(g(c(w)))$, and throughout this process, the service never gets the plaintext w . The main problem here is that we need to have $c^{-1} \circ g \circ c = g$, or equivalently to have g and c commute, which is the necessary condition for a homomorphic cipher. We can extend this notion to a fully homomorphic cipher, one which can generalize to arbitrary computations.

In [1], the authors explore the idea of homomorphic ciphers with respect to the p -adic arithmetic ($+$ and $\cdot$) on the p -adic model of computation. We present their findings here. In section 2, we present the preliminary ideas further, including the definition of homomorphic encryption and the p -adic primitives that are used frequently in the next few sections. In section 3, we address the model of computation that we are working with. In section 4, we provide the main results, namely the (fully) homomorphic ciphers for the p -adic arithmetics, and show that they are the only feasible ones. Finally, in section 5, we show the applications of the main results.

2 Preliminaries

Here we describe the preliminary ideas and notations that are used frequently throughout this exposition. Specifically, we describe the notion of (fully) homomorphic ciphers and encryption and some notations and definitions related to p -adic functions. To review, we define ciphers to be a family of bijective functions of a set M , where these functions are determined by certain parameters. In our case, $M = \mathbb{Z}_p$.

2.1 Homomorphic Encryption Definition

Homomorphic encryption, as its name suggests, is a form of encryption where the cipher's encryption function defines a homomorphism over the operation defined over the plain text group.

Definition 1 (Homomorphic Encryption). Let (X, g_1) be the set X of plain text with operation g_1 . Let (Y, g_2) be the set Y of ciphertext with operation g_2 . Let $f : X \rightarrow Y$ be the encryption function of cipher $\mathcal{C}$. Let m_1 and m_2 be any pair of elements of X . We say $\mathcal{C}$ is a homomorphic encryption if and only if:

$$g_2(f(m_1), f(m_2)) = f(g_1(m_1, m_2))$$

Example 1. [ElGamal Encryption as in [2]]

Alice wants to send Bob a message m . Let $X = (\mathbb{Z}/p\mathbb{Z})^\times$ and $Y = (\mathbb{Z}/p\mathbb{Z})^\times \times (\mathbb{Z}/p\mathbb{Z})^\times$, both with product operation. Let $g, x \in X$ be fixed parameters for all message encryptions, and $r \in X$ be randomly selected every time an encryption is executed. Parameters g and p are public, while x is private and generated by Bob and r is private and generated by Alice.

Bob first sends Alice the value g^x . Alice replies with the ciphertext $(g^r, m(g^x)^r)$. Bob now calculates $(g^r)^x$ and $((g^r)^x)^{-1}$ and retrieves m . Here the encryption function is $f : X \rightarrow Y$ such that $m \mapsto (g^r, m(g^r)^x)$; we will show that it is in fact a homomorphic encryption with respect to multiplication.

Let m_1 and m_2 be two messages that Alice wants to send to Bob with respective random parameters r_1 and r_2 . Their respective ciphertexts at the moment they are encrypted are $f(m_1) = (g^{r_1}, m_1(g^{r_1})^x)$ and $f(m_2) = (g^{r_2}, m_2(g^{r_2})^x)$. Now,

$$f(m_1) \cdot f(m_2) = (g^{r_1+r_2}, (m_1 \cdot m_2)(g^{r_1+r_2})^x)$$

but we know that since r_1 and r_2 are randomly generated so is $r_1 + r_2$, thus

$$f(m_1 \cdot m_2) = (g^{r_1+r_2}, (m_1 \cdot m_2)(g^{r_1+r_2})^x) = f(m_1) \cdot f(m_2)$$

as we wanted. $\square$

Remark It is important to note that because of the way the decryption process works it

really doesn't matter what the random value r is at the moment of encrypting, thus the only requirement we had was for $r_1 + r_2$ to be random.

Now, we want to be able to concatenate different operations that will allow us to have more robust computing capabilities. With this in mind, fully homomorphic encryptions are such that they are fully homomorphic with respect to at least two different operations. In this case we restrict the definition to $X, Y \subseteq \mathbb{Z}_p$.

Definition 2 (Fully Homomorphic Encryption). *Let $X, Y \subseteq \mathbb{Z}_p$ be the sets of plain text and ciphertext. Let $g_1 : \mathbb{Z}_p \times \mathbb{Z}_p \rightarrow \mathbb{Z}_p$ and $g_2 : \mathbb{Z}_p \times \mathbb{Z}_p \rightarrow \mathbb{Z}_p$ be two different operations. Let $f : X \rightarrow Y$ be the encryption function of cipher $\mathcal{C}$. Let m_1 and m_2 be any pair of elements of X . We say the cipher $\mathcal{C}$ is fully homomorphic with respect to g_1 and g_2 if and only if:*

$$f(g_1(m_1, m_2)) = g_1(f(m_1), f(m_2));$$

$$f(g_2(m_1, m_2)) = g_2(f(m_1), f(m_2)).$$

2.2 p -Adic Functions and Dynamics

For any $x \in \mathbb{Z}_p$, if

$$x = x_0 + px_1 + p^2x_2 + \cdots ,$$

for some $x_0, x_1, \dots \in \{0, 1, \dots, p-1\}$, denote for any positive integer k ,

$$(x)_k := x_0 + \cdots + x_{k-1}p^{k-1}, \text{ and}$$

$$[x]_k := (x_0, x_1, \dots, x_{k-1}).$$

We focus on 1-Lipschitz functions that preserve the Haar measure in this exposition, so let us first define 1-Lipschitzness and present some of its properties, then explain what the Haar measure is, and present some theorems about these functions. We will also explore what it means to have logarithms and exponents in $\mathbb{Z}_p$

Definition 3 (Lipschitz). *A continuous function $f : \mathbb{Z}_p \rightarrow \mathbb{Z}_p$ is 1-Lipschitz when*

$$|f(x) - f(y)|_p \leq |x - y|_p, \forall x, y \in \mathbb{Z}_p.$$

Equivalently, for any positive integer k , $x \equiv y \pmod{p^k}$ implies $f(x) \equiv f(y) \pmod{p^k}$.

Let us first explore how we can represent 1-Lipschitz functions. We use the Van der Put series, where for a continuous function $f : \mathbb{Z}_p \rightarrow \mathbb{Z}_p$, there exists a unique $\mathbb{Z}_p$ sequence $B_0, B_1, \dots$ such that

$$f(x) = \sum_{m=0}^{\infty} B_m \chi(m, x),$$

which is the Van der Put representation of the function, with

$$\chi(m, x) = \begin{cases} 1, & \text{if } |x - m|_p \leq p^{-\lfloor \log_p m \rfloor - 1}; \\ 0, & \text{otherwise} \end{cases}, \forall m > 0,$$

and $\chi(0, x) = 1$. The sequence can be computed in the following way,

$$B_m = \begin{cases} f(m) - f(m - m_{\lfloor \log_p m \rfloor} p^{\lfloor \log_p m \rfloor}), & \text{if } m \geq p; \\ f(m), & \text{otherwise} \end{cases}.$$

Therefore, by the definition of a 1-Lipschitz function, there exists a $\mathbb{Z}_p$ sequence $b_0, b_1, \dots$ such that $B_m = p^{\lfloor \log_p m \rfloor} b_m$, so every 1-Lipschitz function f can be represented as

$$f(x) = \sum_{m=0}^{\infty} p^{\lfloor \log_p m \rfloor} b_m \chi(m, x). \quad (1)$$

We discuss the Haar measure next.

Definition 4 (Haar Measure (Informal)). *The Haar measure μ_p is a probability measure, which can be thought of as a probability distribution, on $\mathbb{Z}_p$. With this measure, we have $\mu_p(\mathbb{Z}_p) = 1$ and $\mu_p(B_{p^{-r}}(a)) = p^{-r}$ for any closed ball $B_{p^{-r}}(a)$, with non-negative integer r and $a \in \mathbb{Z}_p$.*

For more detail on this see [3]–[5]. A function f preserves this measure if

$$\mu_p(f^{-1}(U)) = \mu_p(U)$$

for all measurable subsets $U \subset \mathbb{Z}_p$. The following two theorems describe how to represent measure-preserving 1-Lipschitz functions.

Theorem 1 ([3, Theorem 2.1]). *A function f is 1-Lipschitz and measure-preserving if and only if the functions $f_k \equiv f \pmod{p^k}$ is bijective on $\mathbb{Z}/p^k\mathbb{Z}$ for any positive integer k .*

Theorem 2 ([6]). *A 1-Lipschitz function f preserves the measure if and only if in the Van der Put representation in (1) satisfies*

1. $\{b_0, \dots, b_{p-1}\}$ is a complete set of residues modulo p ;
2. For any positive integer k and $m \in \{0, 1, \dots, p^k - 1\}$, $\{b_{m+p^k}, \dots, b_{m+(p-1)p^k}\}$ is a complete set of non-zero residues modulo p .

Finally, we go over the logarithms and exponents in $\mathbb{Z}_p$. The p -adic exponent is the function $\text{EXP}_p : p\mathbb{Z}_p \rightarrow 1 + p\mathbb{Z}_p$, where

$$\text{EXP}_p(x) = \sum_{n=0}^{\infty} \frac{x^n}{n!}, \forall x \in p\mathbb{Z}_p.$$

The sum above converges in $\mathbb{Z}_p$ whenever $x \in p\mathbb{Z}_p$. While the p -adic logarithm is the function $\text{LN}_p : 1 + p\mathbb{Z}_p \rightarrow p\mathbb{Z}_p$ such that $\text{LN}_p = \text{EXP}_p^{-1}$. More precisely,

$$\text{LN}_p(1 + x) = \sum_{n=1}^{\infty} \frac{x^n (-1)^{n+1}}{n}, \forall x \in p\mathbb{Z}_p.$$

3 Models

This section focuses on explaining the automata encryption model and ciphers. We first define the automata that we are working with,

Definition 5 ([1, Section 2]). Let $\mathfrak{A} = \langle \mathcal{T}, \mathcal{S}, \mathcal{L}, S, L \rangle$ be an automaton if $\mathcal{T}$ (the input alphabet) is a finite set of symbols, $\mathcal{L}$ is a finite set of output symbols (output alphabet), $\mathcal{S}$ is a set of states (can be infinite), where $S : \mathcal{T} \times \mathcal{S} \rightarrow \mathcal{S}$ is a state transition function and $L : \mathcal{T} \times \mathcal{S} \rightarrow \mathcal{L}$ is an output function.

The automaton will convert words from the alphabet $\mathcal{T}$ into the alphabet $\mathcal{L}$ using the following: Assume that at the beginning of this process $\mathfrak{A}$ is in the state $s_k \in \mathcal{S}$ for a discrete time $k \geq 0$. Then it processes the input $t_k \in \mathcal{T}$, after this $\mathfrak{A}$ changes its state into $s_{k+1} = S(t_k, s_k)$, and it outputs $l_k = L(t_k, s_k)$. A natural way to extend this type of automaton is to add a specific starting state for the automaton.

Definition 6 ([1, Section 2]). We will call an automaton with a fixed initial state as the initial automaton $\mathfrak{A}_{s_0} = \langle \mathcal{T}, \mathcal{S}, \mathcal{L}, S, L, s_0 \rangle$ is an automaton where the initial state is fixed into s_0 . We denote the function $f_{\mathfrak{A}_{s_0}}$ from words from the alphabet $\mathcal{T}$ to words from the alphabet $\mathcal{L}$ as the function that computes this automaton. In other words, the automaton $\mathfrak{A}_{s_0}$ maps w to $f_{\mathfrak{A}_{s_0}}(w)$ for any word w with alphabet $\mathcal{T}$. In particular, it maps from the words of length n in $\mathcal{T}$, denoted $\mathcal{T}_n$ to words of length n in $\mathcal{L}$, denoted $\mathcal{L}_n$. The function $f_{\mathfrak{A}_{s_0}}$ restricted to $\mathcal{T}_n$ is denoted as $f_{\mathfrak{A}_{s_0}, n} : \mathcal{T}_n \rightarrow \mathcal{L}_n$,

An example from [7] is called the 2-adic adding machine $\mathfrak{A}_1 = \langle \mathbb{F}_2, \mathbb{F}_2, \mathbb{F}_2, S, O, 1 \rangle$, where $S(\chi, s) = \chi s$, $O(\chi, s) = \chi + s$, for $s \in \mathcal{S} = \mathbb{F}_2$ and $\chi \in \mathcal{T} = \mathbb{F}_2$. This automaton simply adds 1 to the input, or more formally, $f_{\mathfrak{A}_1}(x) = x + 1$ for all p -adic integers x .

It would be hard to work directly with functions that can be computed by these automata, but there is a theorem that shows the sufficient and necessary conditions for a function to be computable by these automata, as shown below.

Theorem 3 ([7], Theorem 3). The automaton function $f_{\mathfrak{A}_{s_0}} : \mathbb{Z}_p \rightarrow \mathbb{Z}_p$ of the automaton $\mathfrak{A}_{s_0} = \langle \mathcal{T}, \mathcal{S}, \mathcal{L}, S, L, s_0 \rangle$, $\mathcal{T} = \mathcal{L} = \{0, 1, \dots, p-1\}$ is 1-Lipschitz. Conversely, for every 1-Lipschitz function $f : \mathbb{Z}_p \rightarrow \mathbb{Z}_p$ there exists an automaton $\mathfrak{A}_{s_0} = \langle \{0, 1, \dots, p-1\}, \mathcal{S}, \{0, 1, \dots, p-1\}, S, L, s_0 \rangle$ such that $f = f_{\mathfrak{A}_{s_0}}$.

We will consider on the automaton such that the induced functions $f_{\mathfrak{A}_{s_0}}$ are bijective over the

set of length n words, in other words, $f_{\mathfrak{A}_{s_0}, n} \equiv f_{\mathfrak{A}_{s_0}} \pmod{p^n}$ is bijective on $\mathbb{Z}/p^n\mathbb{Z}$. Then, by Theorem 1, the 1-Lipschitz function $f_{\mathfrak{A}_{s_0}}$ preserves the measure. In this case, by Proposition 4.26 from [2] the function $f_{\mathfrak{A}_{s_0}}$ is bijective on $\mathbb{Z}_p$.

Throughout the paper, the words from alphabet $\mathcal{T}$ to be plain text, and from alphabet $\mathcal{L}$ to be ciphertext. Now notice that the family of these bijective functions f_{s_0} define a cipher with s_0 as an encryption key.

4 p -Adic Arithmetic Homomorphic Ciphers

Now we know that the problem of describing homomorphic ciphers, not just encryptions, is reduced to finding all the functions $f : \mathbb{Z}_p \rightarrow \mathbb{Z}_p$ such that:

1. f is a 1-Lipschitz function;
2. f preserves the measure, and;
3. f defines a homomorphism with respect to the operation we are working on.

So we will focus on finding all such functions for a specific set of operations from now on.

4.1 Existence Homomorphic Ciphers

In this section we will describe which functions define homomorphic ciphers for the arithmetic addition ("+") and arithmetic multiplication ("·").

Theorem 4. *Let $f : \mathbb{Z}_p \rightarrow \mathbb{Z}_p$ be a 1-Lipschitz function, which differs from identically zero and identically one functions. Then*

1. *f defines a homomorphism with respect to the arithmetic addition, i.e., $f(x + y) = f(x) + f(y)$, $x, y \in \mathbb{Z}_p$ if and only if $f(x) = Ax$, $A \in \mathbb{Z}_p$, $A \neq 0$. Such a function f preserves the measure if and only if $A \not\equiv 0 \pmod{p}$;*
2. *f defines a function that respects arithmetic multiplication, i.e., $f(x \cdot y) = f(x) \cdot f(y)$.*

$f(y)$, $x, y \in \mathbb{Z}_p$ if and only if

$$f(x) = \begin{cases} p^k A^k t_0^s \cdot (1 + pt)^a, & \text{if } x = p^k(t_0)(1 + tp), \\ 0, & \text{if } x = 0 \end{cases}$$

where $t_0^{p-1} - 1 = 0$, $A, t, a \in \mathbb{Z}_p$, $s \in \{1, \dots, p-1\}$, $k \geq 0$. Such a function f preserves the measure if and only if $A, a \not\equiv 0 \pmod{p}$, $\gcd(s, p-1) = 1$.

Proof. Lets first consider the case in which f defines a homomorphism with respect to the addition. Then $f(m) = m \cdot f(1)$ for $m \in \mathbb{Z}$. Let $f(1) = A \in \mathbb{Z}_p$ ($A \neq 0$ since we are working with the nonzero functions). Note that using the p -adic metric we say that a function g is continuous over the p -adic integers if $\mathbb{Z}_p$ where continuity means that for all $N > 0$ there exists $M > 0$ such that $|x - y|_p < p^{-M}$ implies $|g(x) - g(y)|_p < p^{-N}$ [8]. Notice that we can use the definition of 1-Lipschitz functions to say that they are continuous by taking $M = N$. Therefore we know that f is continuous over $\mathbb{Z}_p$ and that it is equal to $A \cdot x$ over $\mathbb{Z}$ which is dense over $\mathbb{Z}_p$, thus $f(x) = A \cdot x$ for all $x \in \mathbb{Z}_p$. Functions f preserve the measure if and only if A is invertible, i.e., A is a unit, which is equivalent to saying $A \not\equiv 0 \pmod{p}$.

The following lemma will be useful later:

Lemma 1. *Let T be the set of solutions to the equation $x^{p-1} - 1 = 0$ in $\mathbb{Z}_p$. Let $f_0 : T \rightarrow T$ define a nonzero homomorphism with respect to the multiplication. Then $f_0(x) = x^s$ for some $s \in \{0, 1, \dots, p-2\}$.*

Proof. Firstly, we know that T is the set of all $p-1^{th}$ roots of unit. We further know that $f(1) = 1$, moreover, $f(g) = g^s$ for some s where g is a generator of T , thus $f(x) = f(g^\alpha) = (g^s)^\alpha = (g^\alpha)^s = x^s$ for $x \in T$. Furthermore, $s \in \{0, 1, \dots, p-2\}$ since $p-1$ is the order of $g \pmod{p}$. f is homomorphic since

$$f(a \cdot b) = f(g^\alpha \cdot g^\beta) = f(g^{\alpha+\beta}) = (g^{\alpha+\beta})^s = (g^s)^{\alpha+\beta} = f(a) \cdot f(b)$$

□

We will now consider the case in which f defines a homomorphism with respect to the multiplication. Then, since f is a homomorphism and a nonzero function, we know that $f(0) = 0$ and $f(1) = 1$. Since f is a 1-Lipschitz function we know that

$$|p - 0|_p \geq |f(p) - f(0)|_p,$$

$$|1 + pt - 1|_p \geq |f(1 + pt) - f(1)|_p$$

thus $f(p) \equiv 0 \pmod{p}$ and $f(1 + pt) \equiv 1 \pmod{p}$ for all $t \in \mathbb{Z}_p$. Let $f(p) = pA$, $A \in \mathbb{Z}_p$.

We can now write $x \neq 0$ as $x = p^r t_0(1 + pt)$ with $t \in \mathbb{Z}_p$, $r \geq 0$ and $t_0 \in T$ where T is the set of solutions of $x^{p-1} - 1 = 0$. Since f is a homomorphism we know that $f(t_0)^{p-1} = f(1) = 1$ thus the restriction of f to T is a homomorphism $f_0 : T \rightarrow T$. Since f is a homomorphism so is f_0 , therefore Lemma 1. tells us that $f_0(x) = x^s$ for some $s \in \{0, 1, \dots, p-2\}$. Then the function f can be represented as

$$f(p^r t_0(1 + pt)) = f(p^r) \cdot f(t_0) \cdot f(1 + pt) = (A \cdot p)^r \cdot f_0(t_0) \cdot \phi(1 + pt)$$

where $\phi : 1 + p\mathbb{Z}_p \rightarrow 1 + p\mathbb{Z}_p$ is a 1-Lipschitz function that defines a homomorphism with respect to multiplication. Hence we can write f as

$$f(x) = \begin{cases} A^r \cdot p^r \cdot t_0^s \cdot \phi(1 + pt), & \text{if } x = p^r(t_0)(1 + pt), k \geq 0 \\ 0, & \text{if } x = 0 \end{cases}$$

with $t_0 \in \{1, 2, \dots, p-1\}$, $t \in \mathbb{Z}_p$ and ϕ as defined before.

We will now find the representation of this function ϕ . Recall the functions EXP_p and LN_p defined in section 2.2. Let $g : p\mathbb{Z}_p \rightarrow p\mathbb{Z}_p$ be such that $g(\tau) = \text{LN}_p(\phi(\text{EXP}_p(\tau)))$, then g defines a homomorphism with respect to addition on $p\mathbb{Z}_p$:

$$\begin{aligned}
g(\tau_1 + \tau_2) &= \text{LN}_p(\phi(\text{EXP}_p(\tau_1 + \tau_2))) = \text{LN}_p(\phi(\text{EXP}_p(\tau_1) \cdot \text{EXP}_p(\tau_2))) \\
&= \text{LN}_p(\phi(\text{EXP}_p(\tau_1)) \cdot \phi(\text{EXP}_p(\tau_2))) = \text{LN}_p(\phi(\text{EXP}_p(\tau_1))) + \text{LN}(\phi(\text{EXP}_p(\tau_2))) = g(\tau_1) + g(\tau_2)
\end{aligned}$$

Therefore, there exists $a \in \mathbb{Z}_p$ such that $g(\tau) = a\tau$. This is true since we can use the same arguments we used for part 1 of this theorem 4 by factoring out a p . Now, recall that $\text{EXP}_p(\text{LN}_p(1 + pz)) = 1 + pz, z \in \mathbb{Z}_p$, then

$$\phi(\text{EXP}_p(\tau)) = \text{EXP}_p(g(\tau)) = \text{EXP}_p(a \cdot \tau) = \text{EXP}_p(\tau)^a.$$

Let $y = 1 + pt = \text{EXP}_p(\tau), \tau \in p\mathbb{Z}_p$. Then $\phi(y) = y^a, a \in \mathbb{Z}_p$. Thus, the function f has to be of the form

$$f(x) = f(p^r t_0(1 + pt)) = A^r \cdot p^r \cdot t_0^s \cdot (1 + pt)^a$$

just how we wanted. We can see that functions of this type define homomorphisms on $\mathbb{Z}_p$ with respect to multiplication by checking each of the different calculations for $x = p^r \cdot t_0 \cdot (1 + pt)$.

Let's now find the values of $A, a \in \mathbb{Z}_p$ and $s \in \{1, 2, \dots, p-1\}$ such that our homomorphic function with respect to multiplication f preserves the measure. For this matter we will use theorem 2. Let us find the coefficients of the Van der Put series for f . First we know that $b_{t_0(\text{mod } p)} \equiv f(t_0)(\text{mod } p)$ for $t_0 \in T$, thus the first condition of theorem 2 is satisfied if and only if $T^s = T$, where T^s is the set of s -powers of elements of T , and $T^s = T$ if and only if $\gcd(s, p-1) = 1$.

Now we check the second condition [9]. Let $t \in \{0, 1, \dots, p^r - 1\}, h \in \{1, 2, \dots, p-1\}, t_0 \in T$ and $k \geq 0$. Then Let $m \in \{0, 1, \dots, p^{k+r} - 1\}$ such that $m = t_0 p^k + t p^{k+1}$ with

$t_0 \in T, t \in \{0, 1, \dots, p^{r-1} - 1\}, k \geq 0, r \geq 1$. Let $h \in \{1, 2, \dots, p-1\}$, then we get that

$$\begin{aligned}
b_{p^k t_0(1+p(t+p^r h)) \pmod{p^{k+r+1}}} &= \frac{1}{p^{k+r}} B_{p^k t_0(1+p(t+p^r h)) \pmod{p^{k+r+1}}} \\
&\equiv \frac{1}{p^{k+r}} (f(p^k t_0(1+p(t+p^r h))) - f(p^k t_0(1+pt)) \pmod{p}) \\
&\equiv \frac{1}{p^{k+r}} (p^k \cdot A^k \cdot t_0^s ((1+pt+p^r h)^a - (1+pt)^a) \pmod{p}) \\
&\equiv \frac{1}{p^r} (A^k \cdot t_0^s ((1+pt+p^r h)^a - (1+pt)^a) \pmod{p})
\end{aligned}$$

So right now we are interested in $((1+pt+p^r h)^a - (1+pt)^a) \pmod{p^{r+1}}$ since this will tell us what is $b_{p^k t_0(1+p(t+p^r h)) \pmod{p^{k+r+1}}} \pmod{p}$. To get this value we can use the definitions of EXP_p and LN_p to get

$$\begin{aligned}
(1+pt+p^r h)^a &= \sum_{n=0}^{\infty} \frac{\left(a \cdot \sum_{i=1}^{\infty} \frac{(pt+p^r h)^i (-1)^{i+1}}{i}\right)^n}{n!} \\
&\equiv \sum_{n=0}^{\infty} \frac{\left(a \cdot \frac{pt+p^r h}{1} + a \cdot \sum_{i=2}^{\infty} \frac{(pt)^i (-1)^{i+1}}{i}\right)^n}{n!} \pmod{p^{r+1}} \\
&\equiv \sum_{n=0}^{\infty} \frac{\left(a \cdot p^r h + a \cdot \sum_{i=1}^{\infty} \frac{(pt)^i (-1)^{i+1}}{i}\right)^n}{n!} \pmod{p^{r+1}} \\
&\equiv ap^r h + \sum_{n=0}^{\infty} \frac{\left(a \cdot \sum_{i=1}^{\infty} \frac{(pt)^i (-1)^{i+1}}{i}\right)^n}{n!} \pmod{p^{r+1}} \\
&\equiv ap^r h + (1+pt)^a \pmod{p^{r+1}}
\end{aligned}$$

$$\implies b_{p^k t_0(1+p(t+p^r h)) \pmod{p^{k+r+1}}} \equiv A^k \cdot t_0^s \cdot a \cdot h \pmod{p}$$

Therefore $\{b_{p^k t_0(1+p(t+p^r h)) \pmod{p^{k+r+1}}} : h = 1, 2, \dots, p-1\}$ is the set of all non-zero residues modulo p if and only if $a \not\equiv 0 \pmod{p}$ and $A \not\equiv 0 \pmod{p}$. Then, by theorem 2 the homomorphic function with respect to multiplication f preserves the measure if and only if:

$$\gcd(s, p-1) = 1; a \not\equiv 0 \pmod{p}; A \not\equiv 0 \pmod{p};$$

□

4.2 Requisites for Fully Homomorphic Ciphers

We present one case in which there are no non-trivial fully homomorphic ciphers and another where there are such ciphers. To simplify our analysis, let $\mathcal{H}(\ast)$ be the set of 1-Lipschitz homomorphism with respect to the operation $\ast$ that preserves the measure. Next, we show that there are no non-trivial fully homomorphic ciphers with respect to the pair of operations $+$ and $\cdot$.

Theorem 5.

$$\mathcal{H}(+) \cap \mathcal{H}(\cdot) = \text{Id}.$$

Proof. Let $f \in \mathcal{H}(+) \cap \mathcal{H}(\cdot)$, then because $f \in \mathcal{H}(\cdot)$, then $f(1) = 1$. By Theorem 4, there exists some $A \in \mathbb{Z}_p$ such that $f(x) = Ax$ for all $x \in \mathbb{Z}_p$, so $1 = f(1) = A$, and $f(x) = x$ for all $x \in \mathbb{Z}_p$. Thus, $f = \text{Id}$. $\square$

This means that there is no practical fully homomorphic encryption on the p -adic numbers with respect to both addition and multiplication. Despite this, there could still be pairs of operations that can have some fully homomorphic encryption with respect to them, so we show in the following the operations G such that $+$ and G have such encryptions. For the purposes of this project, we consider only the binary operation $G : \mathbb{Z}_p^2 \rightarrow \mathbb{Z}_p$ that are converging power series,

$$G(x, y) = c + ax + by + \sum_{k \in \mathcal{N}_G} \sum_{i+j=k} c_{i,j} x^i y^j, \quad (2)$$

for some $a, b, c, c_{i,j} \in \mathbb{Z}_p$ and $\mathcal{N}_G \subseteq \mathbb{N}_{>1}$. Furthermore, for any $k \in \mathcal{N}_G$, there exists some $i \geq 0$ and $j \geq 0$ such that $i + j = k$ and $c_{i,j} \neq 0$.

Theorem 6. *Let $f : \mathbb{Z}_p \rightarrow \mathbb{Z}_p$ be a non-identity linear function with $f : x \mapsto Ax$ for some $A \in \mathbb{Z}_p$, $A \neq 0$. f defines a homomorphism to G if and only if $G(x, y) = ax + by$ for some $a, b \in \mathbb{Z}_p$, or $\mathcal{N}_G \neq \emptyset$ and*

1. $c = 0$,

2. $A^d = 1$, where d is the greatest common divisor of the elements in the set $\{k-1 | k \in \mathcal{N}_G\}$.

Proof. Since f is a non-identity linear function, $A \neq 1$. If f is a homomorphism for G , then $AG(x, y) = G(Ax, Ay)$ for all $x, y \in \mathbb{Z}_p$. With $x = y = 0$, we get $Ac = c$, so $c = 0$ because $A \neq 1$. Using the degree $k \in \mathcal{N}_G$ part of G as defined in Equation (2),

$$A \sum_{i+j=k} c_{i,j} x^i y^j = \sum_{i+j=k} c_{i,j} (Ax)^i (Ay)^j = A^k \sum_{i+j=k} c_{i,j} x^i y^j,$$

for all $x, y \in \mathbb{Z}_p$, so $A = A^k$ for all $k \in \mathcal{N}_G$, which means that A satisfies the following system of equation,

$$A^{k-1} = 1, \forall k \in \mathcal{N}_G.$$

In summary, f being the desired function implies that $c = 0$ and the above system of equation holding. The reverse direction can also be shown to be true and it uses a similar line of reasoning.

If $\mathcal{N}_G = \emptyset$, then f is the desired function if and only if $G(x, y) = ax + by$ for some $a, b \in \mathbb{Z}_p$. If $\mathcal{N}_G \neq \emptyset$, then the above system of equation for A is has the same set of solution as $A^d = 1$ with d being the GCD of $\{k-1 | k \in \mathcal{N}_G\}$, which proves this theorem for the case of $\mathcal{N}_G \neq \emptyset$. $\square$

Theorem 6 shows that there are certain operations G with non-trivial fully homomorphic ciphers preserving both addition and G . Furthermore, we can find exactly how many such ciphers exist for a specific G .

When $p \nmid d$, we have that $A^d = 1$ has $\gcd(p-1, d)$ solutions including $A = 1$ by Theorem 3.24 of [10]. When $d = p^k$ for some non-negative integer k , we have $A = 1$ if $p \neq 2$ or $A = \pm 1$ if $p = 2$ by Theorem 18.9 in [11]. Therefore, if we express $d = p^{v_p(d)} n$, there are $\gcd(n, p-1) - 1$ such non-trivial ciphers if $p \neq 2$. If $p = 2$, there is one non-trivial cipher if $v_2(d) \geq 1$, or zero otherwise.

The following are some examples of operations G and p -adic numbers A such that the mapping $f : x \mapsto Ax$ is a homomorphism on $+$ and G that preserves the measure:

1. $G(x, y) = ax + by$, $a, b \in \mathbb{Z}_p$ for any $A \in \mathbb{Z}_p$ such that $p \nmid A$;
2. $G(x, y) = xy^{p-1}$ or $G(x, y) = x^{p-1}y + xy^{p-1}$ with A satisfying $A^{p-1} = 1$;
3. For $p \neq 2$, $G(x, y) = (xy)^{\frac{p-1}{2}}$ with A satisfying $A^{p-1} = 1$;
4. $G(x, y) = \sum_{s=0}^{\infty} p^s (x^{(p-1)s+1} + y^{(p-1)s+1}) = \frac{x}{1-px^{p-1}} + \frac{y}{1-py^{p-1}}$ with A satisfying $A^{p-1} = 1$.

5 Applications

It was proven in the previous section that there are no fully homomorphic ciphers in a p -adic model of encryption, with arithmetic operations. Consider some equation that can be implemented by a computer, using arithmetic operations. As it was proven, there would not exist any fully homomorphic cipher, however it is possible to construct a new fully homomorphic cipher, with new operations over $\mathbb{Z}_p$.

First, consider that in the p -adic model of encryption, with a formula being applied on the original data, using two operations $g_1 : \mathbb{Z}_p \times \mathbb{Z}_p \mapsto \mathbb{Z}_p$, and $g_2 : \mathbb{Z}_p \times \mathbb{Z}_p \mapsto \mathbb{Z}_p$. We will define formulas with respect to g_1 and g_2 over $\mathbb{Z}_p$ as the following:

1. variables being operated with g_1 and g_2 .
2. if F_1, F_2 are formulas, then $g_1(F_1, F_2)$, and $g_2(F_1, F_2)$ are both formulas

The set of formulas with g_1 and g_2 are denoted by $[g_1, g_2]$. For example, the formula $xy + x + z$ is in the set of formulas $[+, \cdot]$

Theorem 7. *Let operations $g_1 : \mathbb{Z}_p \times \mathbb{Z}_p \mapsto \mathbb{Z}_p$, and $g_2 : \mathbb{Z}_p \times \mathbb{Z}_p \mapsto \mathbb{Z}_p$, be defined by formulas from $[+, \cdot]$. Let 1-Lipschitz function $f : \mathbb{Z}_p \mapsto \mathbb{Z}_p$ define a nontrivial homomorphism with respect to the operations g_1 and g_2 . Then $[g_1, g_2] \subset [+, \cdot]$ and $[g_1, g_2] \neq [+, \cdot]$*

Proof. Because $g_1, g_2 \in [+, \cdot]$, which implies that $[g_1, g_2] \subset [+, \cdot]$. Now for the second part, we will continue by contradiction, assuming that $[g_1, g_2] = [+, \cdot]$.

Then “+”, and “.” are defined by $\Psi_+(x_1, x_2)$ and $\Psi.(x_1, x_2)$, for g_1 and g_2 , because f is a homomorphism, then you get the following relationships:

$$f(x_1 + x_2) = f(\Psi_+(x_1, x_2)) = \Psi_+(f(x_1), f(x_2)) = f(x_1) + f(x_2)$$

$$f(x_1 \cdot x_2) = f(\Psi.(x_1, x_2)) = \Psi.(f(x_1), f(x_2)) = f(x_1) \cdot f(x_2)$$

Meaning that f is a homomorphism with respect to “+” and “.”. Then by theorem 5, $\mathcal{H}(+) \cap \mathcal{H}(\cdot) = I$, it follows that f is the identity function, which is a contradiction. $\square$

Based on these results, you can notice that if there exists some fully homomorphic cipher with respect to some operations g_1 and g_2 , with both being formulas based on addition and multiplication, then the cipher is not applicable for all the formulas in [“+”, “.”], since we just proved that $[g_1, g_2] \neq [“+”, “.”]$. Based on this result, we will show a method to approach it using a fully homomorphic encryption to secure remote computing within the p -adic model of encryption.

Let $W(d_1, \dots, d_n)$ be a given formula (or program) by which cloud computing will be performed, where the $d_i \in \mathbb{Z}_p$ are the data. Assume the formula is given in the basis of “+”, and “.”, so now we would want to find new operations g_1 and g_2 to write it as basis. We would want them to be such that $W \in [g_1, g_2]$, and that there exists a fully holomorphic cipher f_a with respect to g_1, g_2 where a is the key and $[g_1, g_2] \neq [“+”, “.”]$

With these conditions, we have that $W(f_a(d_1), \dots, f_a(d_n)) = f_a(W(d_1, \dots, d_n))$. This is what we wanted for it to be able to make the given formula or program secure for cloud computing.

Example 2 ([1])

Let W be the formula we want to perform in our cloud computing. Over 3 variables $x, y, z \in \mathbb{Z}_p$, such that:

$$W(x, y, z) = x^{p-1}y^{(p-1)^2}z + x^{p-1}y^{p-1}z + x^py^{p(p-1)}(1 + y^{p^2-3p+2})$$

Now for our new operators, let us use the same “+” operation in $\mathbb{Z}_p$ and a new operation “*”, such that $a * b = ab^{p-1}$. After writing our formula over our new operations we get:

$$W(x, y, z) = z * (x * y) + (z * x) * y + (x * x) * (y * y) + x * ((x * y) * y)$$

Now from what was already proven in Theorem 6, we can define functions $f : \mathbb{Z}_p \mapsto \mathbb{Z}_p$ such that $f(x) = Ax$, where $A \neq 0$ and $A^{p-1} = 1$, the functions are homomorphisms with respect to “+” and the new operation “*”. Meaning that this family of functions are fully homomorphic ciphers with respect to those new operations, and also with respect to W .

6 Conclusion

We have shown that one can make homomorphic ciphers for any arithmetic over the p -adic integers, which allows us to do computations privately on third-party devices if we shift to the p -adic computation models. However, we also have shown a significant obstacle to making these ciphers practical, namely that there are no non-trivial fully homomorphic ciphers over both p -adic addition and multiplication. This limitation is addressed by providing the possible pairs of operations that have non-trivial fully homomorphic ciphers.

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